DECK 14 · WEEK 4 · AFTERNOON 2

# Fixed Income II Pricing, Yields & the Term Structure

*From a single discount rate to a full spot / par / forward curve*

CFA Level I · Vol 6 Learning Modules 6–9

# Roadmap — Fixed Income II: Pricing, Yields & Term Structure

*4 sections · ~2 hours · Built bottom-up from CFA readings*

**01**

### Bond pricing mechanics

PV of cash flows · discount/par/premium · Greece bond · flat vs full price · matrix pricing

**02**

### Yield spread measures · fixed-rate

current yield · street convention · yield-to-worst · G-spread · Z-spread · OAS intuition

**03**

### Money-market and floater yields

discount rate vs add-on rate · bond-equivalent yield conversion · FRN discount margin

**04**

### Term structure: spot / par / forward

spot curve · par rate · implied forwards · bootstrapping · curve-shape relationships

# Learning objectives

*By the end of this deck you should be able to:*

1. Price a fixed-coupon bond on and between coupon dates, including accrued interest, and classify it as discount, par, or premium.

2. Estimate the price or yield of an illiquid bond using matrix pricing from a ladder of comparable traded bonds.

3. Compute current yield, street-convention YTM, yield-to-call and yield-to-worst on a callable bond, and reconcile rates of different periodicities.

4. Decompose a bond's yield into a benchmark plus G-spread or Z-spread, and sketch how an OAS further adjusts for an embedded option.

5. Convert money-market discount rates to bond-equivalent yields; price a floating-rate note; bootstrap a spot curve from par bond prices and derive par & forward rates.

SECTION 01

# Bond pricing mechanics

# A bond's price is the present value of its promised cash flows discounted at the market discount rate — and moves inversely with that rate

*The market discount rate r is the rate of return investors demand for the bond's credit, liquidity and maturity risks — also called the required yield.*

$$PV = PMT_1 / (1 + r)^1 + PMT_2 / (1 + r)^2 + \ldots + (PMT_n + FV_n) / (1 + r)^n$$

*where: $PMT_t$ = coupon at period t; $FV_n$ = face value at maturity; r = market discount rate per period; n = number of periods to maturity.*

### RATE ↑ → PRICE ↓

Discount factors $(1+r)^t$ grow with r; a higher rate shrinks every future cash flow by more.

### CONVEX, NOT LINEAR

The PV curve is convex in r — a 50-bp cut moves price by more than a 50-bp hike cuts it (second derivative positive).

### LONG BONDS MOVE MORE

The longer the maturity and the lower the coupon, the larger the % price change for a given Δr — duration intuition (Deck 15).

*Source: CFA Institute, Curriculum Vol 6, Learning Module 6, §2 "Bond Pricing and the TVM", pp. 130–134.*

# When coupon < YTM the bond trades at a discount; coupon = YTM →par; coupon > YTM → premium — the three regimes summarised on one line

*BRWA 5-year bond, semi-annual 1.6% periodic coupon (= 3.2% p.a.), priced at three different per-period YTMs.*

| Periodic YTM | Coupon vs rate | Price | Regime | Intuition |
|---|---|---|---|---|
| 1.2% | Coupon > rate | 103.75 | Premium | Excess coupon of +0.4/period for 10 periods, discounted at 1.2% |
| 1.6% | Coupon = rate | 100.00 | Par | Coupon exactly meets the market-demanded rate |
| 2.0% | Coupon < rate | 96.41 | Discount | Deficient coupon of −0.4/period for 10 periods, discounted at |

**PULL-TO-PAR**

A discount (premium) bond's price drifts up (down) toward 100 as maturity approaches, even with the YTM held constant. Total return = coupon income + this predictable capital gain/loss.

**ANNUALIZED STATEMENT**

Equivalently: BRWA is at a premium when stated annual YTM 2.4% < stated annual coupon 3.2%. Interest rates are annualized by multiplying periodic × m.

*Source: CFA Institute, Curriculum Vol 6, LM 6, §2 Example 1 (BRWA bond) and Exhibit 3, pp. 130–133.*

# The Greece 5-year 3.5% coupon bond at a 5% YTM prices at EUR 93.50 per 100 — a worked proof of the discount-bond intuition

*Coupon 3.5% < YTM 5.0% → the bond trades at a discount. Decompose the price into PV-of-coupons plus PV-of-face.*

$$P = C \cdot [\,1 - (1+r)^{-n}\,] / r + FV / (1+r)^n$$

*where: Annuity-form for level coupons + single PV of par. Equivalent to summing each cash flow individually.*

**CFA LM 6 — GREECE 5-YEAR SOVEREIGN · ANNUAL COUPONS**

Face EUR 100, annual coupon 3.5%, market YTM = 5.0%. Cash flows: 3.5, 3.5, 3.5, 3.5, 103.5. Compute price.

**1.** PV of coupons: $C \times [1 - (1.05)^{-5}] / 0.05 = 3.5 \times [1 - 0.78353] / 0.05 = 3.5 \times 4.3295 = \text{EUR } 15.15$
**2.** PV of face: $100 / (1.05)^5 = 100 / 1.27628 = \text{EUR } 78.35$
**3.** Price = PV(coupons) + PV(face) = 15.15 + 78.35 = EUR 93.50
**4.** Check: coupon 3.5% < YTM 5.0% → price < par → discount bond ✓

**P = EUR 93.50 per 100 face (a 6.50% discount to par, pulled to par over 5 years)**

*Source: CFA Institute, Curriculum Vol 6, LM 6, §2 Bond pricing with discount rate, pp. 131–134.*

# Between coupon dates full price = flat price + accrued interest — dealers quote the flat price but the buyer settles on the full price

*If full prices were quoted, investors would see a price jump down every coupon date — a quotation artefact. Flat price removes this sawtooth.*

$$PV_Full = PV_Flat + AI \qquad AI = (t / T) \times PMT$$

*where: t = days elapsed since last coupon; T = days in the full coupon period; PMT = coupon per period. Day count = Act/Act or 30/360.*

### FLAT (CLEAN / QUOTED)

What dealers show on the screen. Removes the mechanical intra-period accrual so price changes reflect only yield moves. Used for all quoted-price statistics and for computing spreads.

### FULL (DIRTY / INVOICE)

Actual cash paid by buyer on settlement. Includes the portion of the next coupon that accrued to the seller during their holding period. AI does NOT depend on yield — only on the calendar.

*Source: CFA Institute, Curriculum Vol 6, LM 6, §3 "Flat Price, Accrued Interest, Full Price", pp. 141–148.*

# Worked — Romania 4.625% annual EUR bond settling 15 Dec 2031 at YTM 3.50% prices at EUR 107.60 full, EUR 3.23 accrued, EUR 104.37 flat

*Previous coupon was 3 April 2031; next coupon 3 April 2032. Settlement 15 Dec 2031 sits in the middle of the coupon period.*

## ROMANIA 4.625% EUR ANNUAL BOND · ACT/ACT DAY COUNT

Maturity 3 Apr 2049. Settlement 15 Dec 2031. YTM on settlement = 3.50%. 17 full coupons remaining.

**1.** Days since last coupon: 3 Apr 2031 → 15 Dec 2031 = 256 days. Days in coupon period = 366 (Act/Act).
**2.** Accrual fraction t/T = 256 / 366 = 0.6995.
**3.** PV as of 3 Apr 2031 (full coupon date): price on-date = $\Sigma\ 4.625 / 1.035^{t} + 100 / 1.035^{17} = 114.69$.
**4.** Full price at 15 Dec: PV_Full = PV_on_date × (1.035)^0.6995 = 114.69 × 1.0244 = EUR 107.60 (incl. AI).

**PV_Full 107.60 · AI = (256/366) × 4.625 = 3.23 · PV_Flat = 107.60 – 3.23 = 104.37**

## WHY THE FLAT PRICE IS THE QUOTED PRICE

On the day after each coupon the quoted price in a stable-yield world is unchanged — accrued interest resets to 0 and the seller's claim on the next coupon is gone. Without this convention, quoted prices would mechanically climb every day between coupons.

*Source: CFA Institute, Curriculum Vol 6, LM 6, §3 Example (Romania eurobond), pp. 145–148.*

# Matrix pricing estimates the yield on an illiquid or new bond by interpolating across comparable traded bonds of similar credit and tenor

*Most bonds trade thinly. When no recent print exists, pricing teams use comparable bonds as a proxy — widely used in dealer quotes.*

**YTM_new ≈ linear interpolation of YTMs of comparable bonds by time-to-maturity**

1. **STEP 1 — Identify comparables**
   Select bonds with similar credit quality, coupon rate, seniority, and (ideally) currency. Recent prints required.

2. **STEP 2 — Compute each comparable YTM**
   Back out YTM for each from observed flat prices using the standard PV equation. Confirm day-count conventions match.

3. **STEP 3 — Interpolate to target tenor**
   Linearly weight the comparable YTMs by maturity distance from the target bond. Apply to target's cash flows to get price.

*Source: CFA Institute, Curriculum Vol 6, LM 6, §4 "Matrix Pricing", pp. 155–163.*

# Worked — price a new 4-year 4.5% annual corporate bond from a 3-year 5.5% at 107.5 and a 5-year 4.5% at 104.75

*Comparables must have similar credit — here, the corporate has just issued two bonds at 3y and 5y, giving a clean ladder for interpolation.*

## MATRIX PRICING A NEW 4-YEAR 4.5% ANNUAL COUPON BOND

3-year 5.50% comparable: price 107.500 → solve YTM. 5-year 4.50% comparable: price 104.750 → solve YTM. Target: 4-year 4.5%.

**1.** Step 1 — 3-year YTM: $107.5 = 5.5/(1+r) + 5.5/(1+r)^2 + 105.5/(1+r)^3 \rightarrow r_3 = 2.608\%$
**2.** Step 2 — 5-year YTM: $104.75 = 4.5/(1+r) + 4.5/(1+r)^2 + 4.5/(1+r)^3 + 4.5/(1+r)^4 + 104.5/(1+r)^5 \rightarrow r_5 = 3.378\%$
**3.** Step 3 — Interpolate to 4y: $r_4 = (r_3 + r_5) / 2 = (2.608\% + 3.378\%) / 2 = 2.993\%$
**4.** Step 4 — Price target: $4.5/(1.02993) + 4.5/(1.02993)^2 + 4.5/(1.02993)^3 + 104.5/(1.02993)^4$
**5.** Step 5 — Sum PVs: 4.370 + 4.244 + 4.121 + 92.815 = 105.55

**Estimated price ≈ EUR 105.55 per 100 face (YTM ≈ 2.99%, implied spread over government benchmark tracked separately)**

*Source: CFA Institute, Curriculum Vol 6, LM 6, §4 matrix-pricing examples, pp. 160–163.*

SECTION 02

# Yield spread measures for fixed-rate bonds

# Current yield, street-convention YTM, true yield and government-equivalent yield are variations on one theme — what we discount and over what day count

*Differences typically measure only a basis point or two — but always material when comparing bonds issued under different conventions.*

| Measure | Definition | Typical use |
|---|---|---|
| Current yield | Annual coupon / flat price — ignores time value, amortization of discount/premium, and reinvestment. | Quick proxy |
| Street-convention YTM | Standard YTM assuming cash flows occur on scheduled dates, ignoring weekends & holidays. | Dealer quotes |
| True yield | YTM using actual payment dates, accounting for weekends/holidays. Always ≤ street-convention yield. | Accounting |
| Government-equivalent yield | Corporate 30/360 yield re-stated on Act/Act to compare with a Treasury benchmark. | G-spread input |
| | Coupon + straight-line amortization of price-to-par / flat | |

**PERIODICITY CONVERSION — NEVER SKIP**

$(1 + APR_m / m)^m = (1 + APR_n / n)^n$ — two bonds with identical cash flows but different compounding conventions produce different YTMs. Align periodicities before computing a spread.

*Source: CFA Institute, Curriculum Vol 6, LM 7, §3 Other yield measures and conventions, pp. 163–168.*

# The yield-to-worst on a callable bond is the minimum of YTM and every yield-to-call — the floor buyers assume when they can't control the option

*6.5% VIVU 7y callable notes at 106.50, semi-annual coupons. Call schedule declines from 103.25 in yr 3-4 to 101.00 in yr 6-7.*

$$P = \Sigma\, PMT / (1+r)^t + \text{Call price} / (1+r)^N \qquad YTW = \min\{ YTC_1, YTC_2, \ldots, YTM \}$$

| Call scenario | Call year | Call price | Semi-annual r | Annualized yield |
|---|---|---|---|---|
| Yield-to-first-call | End of 3 | 103.25 | 2.575% | 5.15% |
| Yield-to-second-call | End of 4 | 102.50 | 2.624% | 5.25% |
| Yield-to-third-call | End of 5 | 101.75 | 2.653% | 5.31% |
| Yield-to-fourth-call | End of 6 | 101.00 | 2.681% | 5.36% |
| Yield-to-maturity (not called) | End of 7 | 100.00 | 2.705% | 5.41% |
| **Yield-to-worst** | – | – | – | **5.15% ($YTC_1$)** |

*Source: CFA Institute, Curriculum Vol 6, LM 7, §3 Example 6 VIVU callable notes, pp. 168–171.*

# A bond's YTM decomposes into a benchmark rate plus an issuer-specific yield spread — macro captured by the benchmark, micro in the spread

*Spread isolates issuer-specific risks (credit, liquidity, tax) from market-wide rate moves. Analysts monitor spreads vs history, not absolute yields.*

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YTM_bond = Benchmark_YTM + Yield spread
G-spread = YTM_bond - interpolated government YTM
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*where: Benchmark = on-the-run sovereign of similar maturity (or interest-rate swap for I-spread). Spread often in bps.*

### G-SPREAD

Difference vs (interpolated) government YTM of matching maturity. Cleanest cross-issuer credit gauge. Single-number.

### I-SPREAD

Difference vs matched-maturity interest-rate swap rate. Common for EUR corporates — swap curve is deeper than the sovereign curve.

### Z-SPREAD

Constant spread added to EVERY benchmark spot rate so PV of cash flows = price. More accurate when the curve is steep.

*Source: CFA Institute, Curriculum Vol 6, LM 7, §4 "Yield Spreads over Benchmark Rates", pp. 171–176.*

# Worked — Russian Federation 5.25% 30y bond at 123.5 six years in has a G-spread of 166 bps over an interpolated 24y Treasury

*At issuance in 2020 the bond priced at 99.625 with a 200-bp spread over the 30y UST. Six years later we compute the current spread.*

**RUSSIA 5.25% USD · SEMI · 30Y ORIGINAL · 24Y REMAINING**

15 Mar 2026 price = 123.5. 20-year UST yields 2.00%; 30-year UST yields 2.25%. Compute current G-spread.

**1.** Step 1 — Current YTM: solve $123.5 = \Sigma\ 2.625/(1+r/2)^t + 100/(1+r/2)^{48}$ → YTM ≈ 3.756%
**2.** Step 2 — Weights for linear interpolation to 24-year tenor:
**3.** $w_{20} = (30 - 24)/(30 - 20) = 60\%$; $w_{30} = 40\%$
**4.** Step 3 — Interpolated 24-year Treasury: $r_{24} = 0.6 \times 2.00\% + 0.4 \times 2.25\% = 2.10\%$
**5.** Step 4 — G-spread = YTM_bond − $r_{24}$ = 3.756% − 2.10% = 1.656%

**G-spread = 166 bps (vs issuance spread of 200 bps → modest credit improvement)**

*Source: CFA Institute, Curriculum Vol 6, LM 7, §4 Example 7 Russian Federation bond, pp. 174–176.*

# The Z-spread is the constant add-on to every benchmark spot rate that makes PV of the bond's cash flows equal its price — more accurate than G-spread when the curve is steep

*G-spread uses ONE discount rate (the YTM). Z-spread uses the whole spot curve plus a constant — the spread an investor "earns at every point".*

$$\text{Price} = \Sigma_t \; PMT_t / (1 + z_t + Z)^t + FV / (1 + z_n + Z)^n$$

$$\text{OAS} = \text{Z-spread} - \text{option value (bps)}$$

*where: $z_t$ = benchmark spot rate for period t; Z = Z-spread (solve numerically). OAS strips out the value of an embedded call / put option.*

### WHY Z > G ON A STEEP CURVE

Early cash flows get a lower discount when using the full curve vs a single 30y YTM — raising their PV, forcing Z > G to re-match the quoted price.

### STATIC / ZERO-VOLATILITY

Z-spread assumes a single deterministic path for rates — no option value. That's why callable-bond analysts pair it with OAS.

### OAS INTUITION

For a callable: Z-spread overstates the credit spread because part of it compensates for the call. OAS = Z – option value = clean credit spread.

*Source: CFA Institute, Curriculum Vol 6, LM 7, §4 Z-spread and OAS, Equations 4–5, pp. 176–178.*

# Worked — Z-spread on a 3-year 5% annual corporate bond priced at 98.50 given spot rates 3.0% / 3.5% / 4.0% is ≈ 160 bps

*Solve by iteration (Excel Goal Seek or bisection). Each cash flow is discounted at its own spot rate plus Z.*

**SOLVE FOR Z SO THAT SUM OF DISCOUNTED CASH FLOWS = 98.50**

Coupons 5, 5, 105 at end of years 1, 2, 3. Spot curve: $z_1$ = 3.0%, $z_2$ = 3.5%, $z_3$ = 4.0%. Quoted price = 98.50.

**1.** Set up: $98.50 = 5 / (1.03 + Z)^1 + 5 / (1.035 + Z)^2 + 105 / (1.04 + Z)^3$
**2.** Trial Z = 1.00%: $5/1.040 + 5/1.045^2 + 105/1.050^3 = 4.808 + 4.580 + 90.702 = 100.09$ (too high)
**3.** Trial Z = 2.00%: $5/1.050 + 5/1.055^2 + 105/1.060^3 = 4.762 + 4.494 + 88.164 = 97.42$ (too low)
**4.** Bisect to Z = 1.60%: $5/1.046 + 5/1.051^2 + 105/1.056^3 = 4.780 + 4.526 + 89.158 = 98.46$
**5.** Refine (Goal Seek): Z ≈ 1.599% so that PV = 98.500 exactly

**Z-spread ≈ 160 bps (constant add-on applied to every spot rate along the curve)**

*Source: CFA Institute, Curriculum Vol 6, LM 7, §4 Z-spread example paraphrased, pp. 176–178.*

SECTION 03

# Money-market and floater yields

# Money-market rates are annualized but not compounded — and come in two mutually inconsistent flavours: discount rate (on FV) and add-on rate (on PV)

*Money-market discount rate UNDERSTATES the true return because it divides earned interest by FV, not PV — a legacy convention.*

$$PV_DR = FV \times [1 - (Days/Year) \times DR] \qquad PV_AOR = FV / [1 + (Days/Year) \times AOR]$$

*where: DR quoted on face, AOR quoted on principal. Day-count: T-bills usually 360; CDs 360 or 365 depending on jurisdiction.*

| Convention | Formula | Typical instruments |
|---|---|---|
| Discount rate (DR) | DR = (Year/Days) × (FV − PV) / FV | T-bills · commercial paper · bankers' acceptances |
| Add-on rate (AOR) | AOR = (Year/Days) × (FV − PV) / PV | Bank CDs · repos · MRR indices |
| Bond-equivalent yield (BEY) | AOR stated on a 365-day year | Standard common basis for comparison |

[1] Convention alone separates these: a Treasury bill and a CD with the same cash flows will quote different numbers.
[2] Always convert to a common basis (usually 365-day add-on = bond-equivalent yield) before comparing.

*Source: CFA Institute, Curriculum Vol 6, LM 8, §3 Money Market Yield Measures, pp. 194–200.*

# Worked — a 90-day CP quoted at 0.100% discount rate on a 360-day year converts to a BEY of 0.102% once re-expressed on a 365-day add-on basis

*Three-step conversion: (1) price from DR; (2) compute AOR on 365-day year; (3) that AOR = BEY.*

**BRWA 90-DAY COMMERCIAL PAPER · DISCOUNT RATE 0.100% · 360-DAY YEAR**

Compute the price, then the BEY, then compare to a 90-day CD quoted at 0.120% AOR on a 365-day year (same credit).

**1.** Step 1 — Price from DR: PV = 100 × [1 − (90/360) × 0.00100] = 100 × 0.99975 = 99.975
**2.** Step 2 — AOR on 365-day year: AOR = (365/90) × (100 − 99.975) / 99.975 = 4.0555 × 0.0002501 = 0.001014
**3.** Step 3 — Express as BEY: BEY = AOR = 0.1014% ≈ 0.102% (365-day basis)
**4.** Compare vs CD: CFP Bank 90-day CD AOR on 365-day = 0.120%. BRWA CP at BEY 0.101% → CD is higher-yielding.
**5.** Interpretation: BRWA appears cheaper at 0.100% DR but only because of the 360-day convention masking 1.4% annualization uplift.

**BRWA BEY = 0.102% vs CFP Bank CD = 0.120% → CD offers the higher return on an apples-to-apples basis**

*Source: CFA Institute, Curriculum Vol 6, LM 8, §3 Example 3 and Exhibit 3, pp. 197–200.*

# A floating-rate note resets its coupon each period to MRR + quoted margin — so it trades at par only when the required margin equals the quoted margin

*At every reset date cash flows "snap" back to market rates, so interest-rate risk is reduced. Residual price risk is driven by CREDIT — the discount margin.*

$$PV = \Sigma \left[ (MRR + QM)/m \times FV \right] / \left[1 + (MRR + DM)/m \right]^{t} + FV / \left[1 + (MRR + DM)/m \right]^{n}$$

*where: QM = quoted margin (locked at issuance); DM = discount margin (market-determined, reflects current credit). Par if QM = DM.*

**QM < DM → DISCOUNT**

Issuer paying "deficient" margin vs what market now demands — bond falls below par. Credit has deteriorated.

**QM = DM → PAR**

Market-required margin = locked quoted margin. Bond prices exactly at 100 on each reset date.

**QM > DM → PREMIUM**

Issuer "overpaying" vs current market. Credit improved since issuance — bond trades above par.

*Source: CFA Institute, Curriculum Vol 6, LM 8, §2 FRN Pricing, Equation 1, pp. 188–193.*

# Worked — Antelas AG 4y FRN paying 3m MRR of −0.55% + 250 bps priced at 97 implies a discount margin of 329 bps — a clear credit downgrade

*Discount is 3 pts below par → market requires more compensation than the 250-bp locked QM. Solve for the required margin implied by price.*

**ANTELAS AG FRN · EUR · QUARTERLY · 4-YEAR · 30/360**

QM = 250 bps; MRR (3m) = −0.55%; Price = 97 per 100 par; N = 16 remaining periods. Face = 100.

**1.** Step 1 — Periodic coupon: PMT = (MRR + QM) / m × FV = (−0.0055 + 0.025) / 4 × 100 = 0.4875
**2.** Step 2 — Solve for periodic rate r using PV = −97, FV = 100, PMT = 0.4875, N = 16 (Excel RATE)
**3.** r = 0.006861 per period
**4.** Step 3 — Back out DM: r = (MRR + DM) / m → 0.006861 = (−0.0055 + DM) / 4
**5.** DM = 4 × 0.006861 + 0.0055 = 0.0329

**DM = 329 bps vs QM of 250 bps → market demands 79 bps MORE than at issuance (credit deterioration)**

*Source: CFA Institute, Curriculum Vol 6, LM 8, §2 Example 1 Antelas AG FRN, pp. 191–194.*

SECTION 04

# Term structure: spot, par & forward

# Spot rates are zero-coupon yields — one rate per maturity — and form the ideal dataset because they carry no coupon reinvestment risk

*In practice, strips (stripped coupon bonds) supply direct spot observations. Elsewhere, analysts bootstrap spots from coupon bonds.*

| Maturity | 1y | 2y | 3y | 5y | 7y | 10y | 20y | 30y |
|---|---|---|---|---|---|---|---|---|
| Canada spot | 0.31 | 0.57 | 0.80 | 1.11 | 1.30 | 1.58 | 1.98 | 2.06 |
| Australia spot | 0.03 | 0.07 | 0.30 | 0.81 | 1.17 | 1.52 | 2.35 | 2.47 |
| Shape | ↗ | ↗ | ↗ | ↗ | ↗ | ↗ | ↗ | → |

**UPWARD-SLOPING (NORMAL)**

Long rates > short rates. Positive maturity premium. Typical in risk-on, moderate-inflation regimes.

**DOWNWARD-SLOPING (INVERTED)**

Long rates < short rates. Usually precedes recession — market expects rate cuts ahead.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §2 "Maturity Structure and Spot Rates", pp. 209–216.*

# A bond's no-arbitrage price is the sum of each cash flow discounted at its OWN spot rate — not at a single YTM

*Using a flat YTM mis-allocates the discount factor across the timeline. Arbitrage arguments force prices to equal the spot-rate PV.*

$$P = PMT_1 / (1 + Z_1)^1 + PMT_2 / (1 + Z_2)^2 + \ldots + (PMT_n + FV) / (1 + Z_n)^n$$

*where: $Z_t$ = spot rate for maturity t. For corporate issuers, add a credit spread (Z-spread) uniformly to every spot.*

### SPOT-PRICING WINS

No-arbitrage. A dealer stripping coupons and pricing each at its own spot must equal whole-bond price — else a riskless profit exists.

### YTM STILL USEFUL

YTM compresses the curve into one number — convenient for quoting, benchmarking, and spread measures. Just know that behind it lives the spot curve.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §2 Bond Pricing with Spot Rates, pp. 213–216.*

# Worked — a 3-year 5% annual bond priced against spot rates 2% / 3% / 4% trades at 102.96 with an implied YTM of 3.94%

*YTM ≠ 5-year spot. It is a weighted mix of all three spots — mostly driven by the 3-year because that's where most of the PV sits.*

**3-YEAR 5% ANNUAL COUPON BOND · PAR 100**

Spot curve: $Z_1 = 2\%$, $Z_2 = 3\%$, $Z_3 = 4\%$. Coupons: 5, 5, 105 at end of years 1-3. Compute price and YTM.

**1.** PV of $C_1$: $5 / (1.02)^1 = 4.902$
**2.** PV of $C_2$: $5 / (1.03)^2 = 4.713$
**3.** PV of $C_3$+Face: $105 / (1.04)^3 = 93.345$
**4.** Price = 4.902 + 4.713 + 93.345 = 102.960 (premium — coupon 5% beats every spot)
**5.** Solve for YTM: $102.96 = 5/(1+r) + 5/(1+r)^2 + 105/(1+r)^3 \rightarrow r = 3.935\%$

**Price = 102.96 · Implied YTM = 3.935% (between $Z_1$ 2% and $Z_3$ 4%, closest to 3-year spot because the final CF dominates)**

*Source: CFA Institute, Curriculum Vol 6, LM 9, §2 numerical illustration, pp. 213–214.*

# A par rate is the coupon that makes a bond price at 100 against the spot curve — the core of the published Treasury yield curve

*US Treasury publishes par rates each day because actual on-the-run bonds rarely price exactly at par — par rates are the clean hypothetical.*

$$100 = \Sigma_t \; PMT / (1 + Z_t)^t + (PMT + 100) / (1 + Z_n)^n \rightarrow \text{solve for } PMT$$

*where: Solve the PV equation for PMT given the spot curve. The resulting PMT / 100 is the par rate per period.*

### PAR RATE vs YTM

YTM depends on a specific bond's coupon; par rate is the SINGLE coupon that produces a par-priced bond. Par rate is purely about the curve.

### CURVE-SHAPE RULES

Upward curve: par rate < spot rate (averaging across earlier, lower spots).
Downward curve: par rate > spot rate.
Flat curve: par rate = spot rate.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §3 "Par Rates from Spot Rates", pp. 217–218.*

# Worked — bootstrap 2-year and 3-year par rates from spot rates 5.263% / 5.616% / 6.359%

*One year: par rate trivially equals $Z_1$. For longer tenors, solve a polynomial in PMT against 100. Shown for 2y and 3y.*

**SPOT RATES GIVEN: $Z_1$ = 5.263%, $Z_2$ = 5.616%, $Z_3$ = 6.359%**

Find the annual coupon PMT that makes each bond price at 100. (Annual compounding, periodicity = 1.)

**1.** 1-year par: 100 = (PMT + 100) / 1.05263 → PMT = 5.263 → par rate = 5.263%
**2.** 2-year par: $100 = PMT/1.05263 + (PMT + 100)/1.05616^2$
**3.** $PMT \times [1/1.05263 + 1/1.05616^2] + 100/1.05616^2 = 100$
**4.** $PMT \times [0.95000 + 0.89635] + 89.635 = 100$ → $PMT \times 1.84635 = 10.365$ → PMT = 5.614 ≈ 5.61%
**5.** 3-year par: $100 = PMT/1.05263 + PMT/1.05616^2 + (PMT+100)/1.06359^3$ → PMT = 6.306%

**1y par = 5.263% · 2y par = 5.606% · 3y par = 6.306% (each below the corresponding spot, curve is upward-sloping)**

*Source: CFA Institute, Curriculum Vol 6, LM 9, §3 par-rate illustration, pp. 217–218.*

# An implied forward rate is the breakeven reinvestment rate that makes a short-then-reinvest strategy equivalent to holding a longer zero

*Naming: "AyBy" = B-year rate starting A years from today. So "3y1y" is the 1-year rate, 3 years from now.*

$$(1 + Z_A)^A \times (1 + IFR_{A,B-A})^{B-A} = (1 + Z_B)^B$$

*where: $Z_A$ = shorter-tenor spot; $Z_B$ = longer-tenor spot; IFR = the implied (B − A)-year forward rate starting at time A.*

### BREAKEVEN INTUITION

If you expect rates HIGHER than IFR, go short (roll). If LOWER, go long (lock). IFR is the indifference point.

### SPOT CURVE → FORWARD CURVE

Each forward rate is a ratio of adjacent spots. $(1+Z_B)^B / (1+Z_A)^A$, then take the (B−A)-th root.

### FORWARD CURVE → SPOT CURVE

Spot = geometric average of forwards: $(1+Z_n)^n = (1+0y1y)(1+1y1y)\ldots(1+(n-1)y1y)$. Three curves, one information set.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §3 "Forward Rates from Spot Rates", pp. 218–220.*

# Worked — the 2y1y implied forward rate from 2y and 3y spots of 3.0% and 4.0% is 6.03% — the breakeven for year 3 reinvestment

*Investing at $Z_3$ = 4% for 3y must equal investing at $Z_2$ = 3% for 2y and then rolling at the 2y1y — solve for the rolled rate.*

## COMPUTE THE 2Y1Y FORWARD RATE

Spot curve: $Z_2$ = 3.0%, $Z_3$ = 4.0%. Annual compounding. Find IFR_{2,1}.

**1.** Start: $(1 + Z_2)^2 \times (1 + \text{IFR_\{2,1\}})^1 = (1 + Z_3)^3$
**2.** $(1.03)^2 \times (1 + \text{IFR_\{2,1\}}) = (1.04)^3$
**3.** $1.0609 \times (1 + \text{IFR_\{2,1\}}) = 1.124864$
**4.** $(1 + \text{IFR_\{2,1\}}) = 1.124864 / 1.0609 = 1.06032 \rightarrow \text{IFR_\{2,1\}} = 6.03\%$

**2y1y = 6.03% (much higher than 4% 3y-spot: market priced into forwards a year-3 rate well above today's level)**

## TRADER TAKE

If your view is the 1-year rate two years from now will be 4.5% (below 6.03%), LOCK-IN: buy the 3-year zero now. If you think it will be 7% (above 6.03%), ROLL: buy the 2-year zero and reinvest at the expected higher rate.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §3 Example 3, pp. 219–220.*

# Spot, par, and forward curves carry identical information — three lenses, one curve; their ordering depends only on the spot curve's slope

*Knowing the spot curve mechanically determines the par and forward curves; knowing any one lets you reconstruct the other two.*

| Curve slope | Forward vs spot | Par vs spot | Economic interpretation |
|---|---|---|---|
| Upward-sloping (normal) | Forward > spot | Par < spot | Market expects higher future rates |
| Flat | Forward = spot | Par = spot | Expectations: rates stay constant |
| Downward-sloping (inverted) | Forward < spot | Par > spot | Market expects lower future rates / recession |

**WHY FORWARDS SIT ABOVE SPOT ON AN UPWARD CURVE**

Long spot = geometric average of shorter spot and forwards. If $Z_3 > Z_2$, then 2y1y must exceed $Z_3$ to drag the average UP. So forwards overshoot the spot curve when it rises.

**MATH IDENTITY**

$(1 + Z_n)^n = \Pi_i (1 + iy1y)$ — spot is the geometric average of one-year forwards.

*Source: CFA Institute, Curriculum Vol 6, LM 9, §4 "Spot, Par, Forward Yield Curves", pp. 225–229.*

# Worked — bootstrap the full 3-year spot curve from par bonds at 2%, 2.5% and 3% coupons — Z = 2.00% / 2.51% / 3.02%

*Bootstrap = solve sequentially. $Z_1$ from the 1-year par bond. $Z_2$ plugs in $Z_1$. $Z_3$ plugs in $Z_1$, $Z_2$. Builds the entire spot curve.*

**PAR BONDS (EACH PRICED AT 100, ANNUAL COUPONS)**

1-year 2% par bond. 2-year 2.5% par bond. 3-year 3% par bond. Recover the spot curve.

**1.** $Z_1$: $100 = 102 / (1 + Z_1) \rightarrow Z_1 = 2.000\%$
**2.** $Z_2$: $100 = 2.5 / 1.02 + 102.5 / (1+Z_2)^2 \rightarrow 97.549 = 102.5 / (1+Z_2)^2 \rightarrow Z_2 = 2.506\%$
**3.** Note $Z_2 > Z_1$ (and > 2y par coupon 2.5%) — upward-sloping curve
**4.** $Z_3$: $100 = 3 / 1.02 + 3 / 1.02506^2 + 103 / (1+Z_3)^3$
**5.** $= 2.941 + 2.855 + 103/(1+Z_3)^3 \rightarrow 94.204 = 103/(1+Z_3)^3 \rightarrow Z_3 = 3.020\%$

**Spot curve: $Z_1$ = 2.00% · $Z_2$ = 2.51% · $Z_3$ = 3.02% (each spot > same-tenor par rate, confirming upward slope)**

*Source: CFA Institute, Curriculum Vol 6, LM 9, §3 bootstrapping synthesis, pp. 218–224.*

KEY TAKEAWAYS

# Deck 14 — Fixed Income II: Pricing, Yields & Term Structure

- Bond price = PV of coupons + PV of face at the market discount rate. Coupon vs YTM determines discount / par / premium; the Greece 5y 3.5% at 5% YTM prices at 93.50.
- Between coupon dates, full price = flat price + accrued interest; dealers quote flat (AI doesn't depend on yield).
- Matrix pricing interpolates comparable-bond YTMs to price illiquid or new issues — three-step process: identify, back out YTMs, interpolate.
- Yield-to-worst = min of YTM and every yield-to-call — the conservative yield on a callable. VIVU worked example: YTW = 5.15%.
- G-spread = YTM − interpolated government YTM. Z-spread = constant add-on to every spot rate. OAS = Z-spread − option value.
- Money-market rates come in DR (T-bills, CP) and AOR (CDs, repos) flavours — always convert to a 365-day bond-equivalent yield before comparing.
- FRN prices at par when quoted margin = discount margin; Antelas AG 4y FRN at 97 implies a DM of 329 bps against a QM of 250 bps → credit widened.
- Spot, par and forward curves carry the same information — bootstrap spots from par bonds, forwards from spots; upward curve ⇒forwards > spot > par.